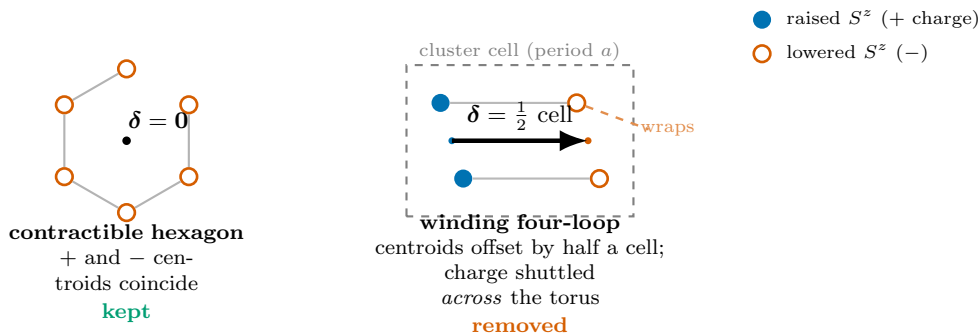


(a) charge transport δ of a ring exchange



(b) the flux must sweep 4π : charge moves by $\frac{a}{2}$

A uniform flux φ gives a process transporting charge δ the phase $e^{-i\varphi \cdot (\mathbf{L}^{-1}\delta)}$. For a winding loop $\mathbf{L}^{-1}\delta = \frac{1}{2}$, so the phase is $e^{-i\varphi/2}$ – it winds at *half* the rate of φ . Averaging kills it only over a *full* turn of the phase, i.e. $\varphi \in [0, 4\pi)$; over $[0, 2\pi)$ the phase makes half a turn and a $\sim 40\%$ residual survives.

